

Mutual Fund Analytics Project — Data Dictionary

Overview

This document defines all datasets, columns, data types, business meanings, and source references used in the Mutual Fund Analytics Data Warehouse project.

1. Dataset: fact_performance

Description

Contains mutual fund performance metrics, risk measures, returns, and AUM information.

Source

- Cleaned from: cleaned_scheme_performance.csv
- Original source: Mutual fund performance dataset

Column Name	Data Type	Business Definition
amfi_code	BIGINT	Unique AMFI scheme identifier
scheme_name	VARCHAR(255)	Official mutual fund scheme name
fund_house	VARCHAR(255)	Asset Management Company (AMC)
category	VARCHAR(100)	Fund category (Equity, Debt, Hybrid, etc.)
plan	VARCHAR(50)	Direct or Regular plan
return_1yr_pct	DECIMAL(6,2)	1-year annualized return percentage
return_3yr_pct	DECIMAL(6,2)	3-year annualized return percentage
return_5yr_pct	DECIMAL(6,2)	5-year annualized return percentage
benchmark_3yr_pct	DECIMAL(6,2)	Benchmark index 3-year return
alpha	DECIMAL(6,2)	Risk-adjusted excess return
beta	DECIMAL(6,2)	Market volatility sensitivity
sharpe_ratio	DECIMAL(6,2)	Risk-adjusted return metric
sortino_ratio	DECIMAL(6,2)	Downside-risk-adjusted return
std_dev_ann_pct	DECIMAL(6,2)	Annualized standard deviation

Column Name	Data Type	Business Definition
max_drawdown_pct	DECIMAL(6,2)	Maximum portfolio decline percentage
aum_crore	DECIMAL(15,2)	Assets Under Management in crores
expense_ratio_pct	DECIMAL(5,2)	Annual management fee percentage
morningstar_rating	INTEGER	Mutual fund rating (1-5)
risk_grade	VARCHAR(50)	Risk category of the fund

2. Dataset: fact_nav

Description

Stores daily Net Asset Value (NAV) history for mutual fund schemes.

Source

- Cleaned from: cleaned_nav_history.csv
- Original source: Historical NAV dataset

Column Name	Data Type	Business Definition
amfi_code	BIGINT	Unique AMFI scheme identifier
date	DATE	NAV reporting date
nav	DECIMAL(10,4)	Net Asset Value of the scheme

3. Dataset: fact_transactions

Description

Contains investor-level transaction history including SIPs, redemptions, and lump sum investments.

Source

- Cleaned from: cleaned_investor_transactions.csv
- Original source: Investor transaction dataset

Column Name	Data Type	Business Definition
investor_id	VARCHAR(50)	Unique investor identifier
transaction_date	DATE	Transaction execution date
amfi_code	BIGINT	Mutual fund scheme identifier
transaction_type	VARCHAR(50)	SIP, Lumpsum, or Redemption
amount_inr	DECIMAL(15,2)	Transaction amount in INR
state	VARCHAR(100)	Investor state
city	VARCHAR(100)	Investor city
city_tier	VARCHAR(20)	T30/B30 city classification
age_group	VARCHAR(20)	Investor age bracket
gender	VARCHAR(20)	Investor gender
annual_income_lakh	DECIMAL(10,2)	Annual income in lakhs
payment_mode	VARCHAR(50)	Payment method used
kyc_status	VARCHAR(20)	Investor KYC verification status

4. Dataset: `dim_fund`

Description

Dimension table containing mutual fund master metadata.

Source

- Fund master dataset

Column Name	Data Type	Business Definition
fund_key	INTEGER	Surrogate warehouse key
amfi_code	BIGINT	AMFI scheme code
scheme_name	VARCHAR(255)	Mutual fund scheme name
fund_house	VARCHAR(255)	Asset Management Company
category	VARCHAR(100)	Primary fund category
sub_category	VARCHAR(100)	Secondary fund category

Column Name	Data Type	Business Definition
plan	VARCHAR(50)	Direct or Regular plan
risk_category	VARCHAR(50)	Fund risk classification
sebi_category_code	VARCHAR(20)	SEBI regulatory category code
benchmark	VARCHAR(255)	Benchmark index
fund_manager	VARCHAR(255)	Fund manager name
launch_date	DATE	Scheme launch date
min_sip_amount	DECIMAL(10,2)	Minimum SIP investment
min_lumpsum_amount	DECIMAL(10,2)	Minimum lump sum investment
expense_ratio_pct	DECIMAL(5,2)	Fund expense ratio
exit_load_pct	DECIMAL(5,2)	Exit load percentage

5. Dataset: dim_date

Description

Date dimension table for time-series analytics.

Column Name	Data Type	Business Definition
date_key	INTEGER	Surrogate date key
full_date	DATE	Calendar date
day_number	INTEGER	Day of month
month_number	INTEGER	Month number
quarter_number	INTEGER	Quarter number
year_number	INTEGER	Year
weekday_name	VARCHAR(20)	Day name
month_name	VARCHAR(20)	Month name

Business Rules

NAV Rules

- NAV must be greater than 0
- Missing NAV values are forward-filled for holidays/weekends

Transaction Rules

- amount_inr > 0
- transaction_type standardized to:
 - SIP
 - Lumpsum
 - Redemption

KYC Rules

Allowed values:

- Verified
- Pending
- Rejected

Expense Ratio Rules

Expected valid range:

- 0.1% to 2.5%
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Warehouse Architecture

Fact Tables

- fact_nav
- fact_transactions
- fact_performance

Dimension Tables

- dim_fund
 - dim_date
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Technology Stack

Component	Technology
Database	MySQL
ETL	Python + Pandas
ORM/Connector	SQLAlchemy
Visualization	Power BI / Matplotlib
Data Cleaning	Pandas
Query Language	SQL

Author

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